

Project Interim Report
On
File Compression Simulator
(Using Core Java & OOP Concepts)

RU-100-01-00012
Object Oriented Programming with Java

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Abstract

Music Playlist Management System

Digital music players rely on efficient playlist management to organize and play songs seamlessly. This project focuses on building a **Music Playlist Management System** using **Object-Oriented Programming (OOP)** concepts

Introduction

In today's digital world, music applications require efficient playlist management systems. This project is designed to simulate such a system using Java and OOP principles.

The Music Playlist Management System allows users to:

- Add songs to a playlist
- View songs in order
- Shuffle songs randomly

Objectives

- To implement OOP concepts in real-world application
- To manage song records efficiently
- To demonstrate aggregation (HAS-A relationship)
- To use arrays of objects

Scope

This project is suitable for small-scale music applications. It can be extended in the future by:

- Adding a database
- Creating GUI (JavaFX/Swing)
- Adding features like delete/search song

System Requirements

Hardware

- Computer with minimum 4GB RAM

Software

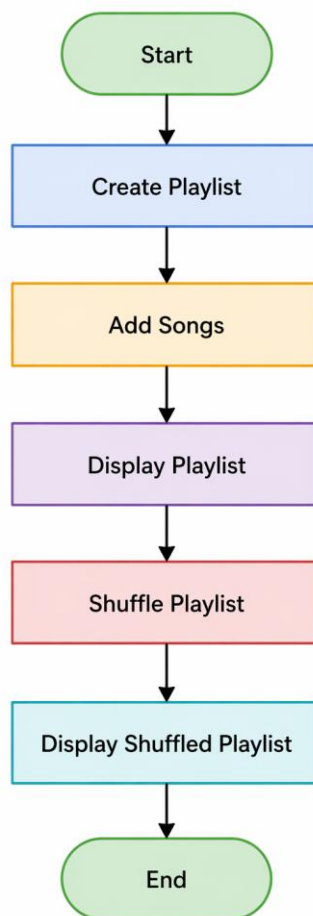
- Java JDK
- IDE (VS Code / Eclipse)

Methodology

The system works in the following steps:

1. Create Song objects with title and artist
2. Add songs to Playlist
3. Store songs in an array
4. Display songs using loop
5. Shuffle playlist using random logic

Flowchart



Class Diagram

Class: Song

- Variables: title, artist
- Methods: constructor, getters

Class: Playlist

- Variables: Song[] songs, int count
- Methods: addSong(), shufflePlaylist(), display()

Implementation

Song Class

Stores song details like title and artist.

Playlist Class

Manages collection of songs using array.

Main Class

Creates playlist

Adds songs

Displays playlist

Shuffles songs

Sample Input / Output

Playlist:

1. Believer - Imagine Dragons
2. Shape of You - Ed Sheeran
3. Blinding Lights - The Weeknd

After Shuffle:

1. Blinding Lights - The Weeknd
2. Believer - Imagine Dragons
3. Shape of You - Ed Sheeran

Future Enhancements

- Add delete song feature
- Add search functionality
- Use database (MySQL)
- Create GUI interface

Gantt Chart

Task	Duration
Planning	2 Days
Design	2 Days
Coding	4 Days
Testing	2 Days

Conclusion

The project successfully demonstrates the use of Object-Oriented Programming concepts in Java. It provides a simple and effective system for managing music playlists using arrays and aggregation.

References

- H. Schildt, *Java: The Complete Reference*, McGraw-Hill, 2018
*
- Oracle Java Documentation – <https://docs.oracle.com>

